

No. of Functional Features Included in the Solution

Date	8 JULY 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication (if implemented)	Allows authorized users to log in securely before accessing the application.	Authentication	Done	High
2	Weather Data Input	Users enter annual rainfall, cloud visibility, and seasonal rainfall details through the web interface.	User Interface	Done	High
3	Data Preprocessing	Cleans and prepares weather data before sending it to the prediction model.	Backend Logic	Done	High
4	Flood Prediction Model	Predicts flood occurrence using Decision Tree, Random Forest, KNN, and XGBoost algorithms.	Machine Learning	Done	High
5	Model Selection	Compares model performance and selects XGBoost as the best-performing model (96.55% accuracy).	Machine Learning	Done	Medium
6	Prediction Result Display	Displays flood risk prediction clearly through the Flask web application.	Frontend / Flask	Done	High
7	Dataset Management	Stores and manages historical weather data used for training and testing the models.	Data Storage	Done	Medium
8	IBM Cloud Deployment Support	Enables future deployment of the application on IBM Cloud for remote accessibility.	Deployment	In Progress	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8

Total Features Implemented	7
Core / Must-Have Features	6 (Weather Data Input, Data Preprocessing, Flood Prediction, Model Selection, Result Display, Dataset Management)
Additional / Nice-to-Have Features	2 (User Authentication, IBM Cloud Deployment)
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Weather Data Input Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, ML Prediction Engine, Model Selection
3	Database / Storage	1	Historical Weather Dataset (CSV)
4	API / Integration	1	IBM Cloud Deployment (<i>or Weather API if integrated</i>)
5	Security / Authentication	1	User Login Authentication (<i>if implemented</i>)